

netdiscover / arp-scan / fping

: Nmap이 포트 스캐닝이라는 주 작업을 시작하기 전에, 효율성과 속도를 높이기 위해 선행되어 사용되는 보조 도구 역할

- 보조 도구 사용의 이점 및 목적

=> 시간 절약 및 패킷 절약 !

도구	Nmap과의 연관 역할	목적	명령어
arp-scan	ARP 기반 호스트 목록 생성	ARP를 사용하여 IP와 MAC 주소 매핑 확인	arp-scan -l (로컬 서브넷 전체) arp-scan [서브넷/CIDR]
netdiscover		ARP를 사용하여 호스트 탐색 및 MAC 정보 수집	netdiscover -r [서브넷/CIDR]
fping	ICMP 기반 호스트 생존 확인	ICMP를 사용하여 가장 빠르게 호스트 생존 여부 확인	fping -g [서브넷 IP 시작] [서브넷 IP 끝]

- 스캐닝 과정

1. 활성 호스트 목록 생성: fping이나 arp-scan을 사용하여 네트워크에서 응답하는 IP 주소 목록(예: 192.168.1.10, 192.168.1.50...)을 생성
2. Nmap 스캔: Nmap을 실행할 때, 모든 IP가 아닌 이 목록에 있는 IP에 대해서만 포트 스캔을 수행하도록 지정합니다.

[실습]

1단계 : Rocky Linux 서버에서 **tcpdump**를 설치하고, 공격이 시작될 때 네트워크 인터페이스를 모니터링합니다.

1. **tcpdump** 설치:

```
sudo dnf install tcpdump -y  
apt install tcpdump -y
```

2. 모니터링 시작: 공격받을 로컬 인터페이스(**eth0** 또는 **ensXX**)를 지정하여 실행합니다. 이제 이 창은 **Kali**에서 패킷이 들어오는 순간을 실시간으로 기록하게 됩니다.

```
sudo tcpdump -i [인터페이스 이름] -n  
**[ip a]를 통해 인터페이스 확인 가능
```

```
예: sudo tcpdump -i ens160 -n icmp  
sudo tcpdump -i ens160 -n arp
```

2단계: Kali Linux 공격 실행

록키에서 TcpDump를 켜놓은 상태에서 Kali Linux 터미널로 세 가지 도구를 순서대로 실행하며 로컬 네트워크(예: 192.168.1.0/24)를 스캔합니다. 스캔할 IP 범위에 **Rocky Linux** 서버의 **IP** 주소가 포함되어야 합니다.

[fping]

칼리 명령어 : `fping -g [시작 IP] [끝 IP]` (`-g` = IP범위를 자동 스캔)

```
└─# fping -g 172.16.27.150 172.16.27.155
172.16.27.154 is alive
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.152
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.152
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.152
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.152
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.151
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.151
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.151
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.151
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.150
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.150
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.150
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.150
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.155
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.155
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.155
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.153
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.153
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.153
ICMP Host Unreachable from 172.16.27.50 for ICMP Echo sent to 172.16.27.153
172.16.27.150 is unreachable
172.16.27.151 is unreachable
172.16.27.152 is unreachable
172.16.27.153 is unreachable
172.16.27.155 is unreachable
```

록키 명령어 : `tcpdump -i ens160 -n icmp`

```
0 packets dropped by kernel
[root@localhost ~]# tcpdump -i ens160 -n icmp
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens160, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:34:02.448163 IP 172.16.27.50 > 172.16.27.154: ICMP echo request, id 2331, seq 4, length 64
13:34:02.448186 IP 172.16.27.154 > 172.16.27.50: ICMP echo reply, id 2331, seq 4, length 64
13:34:39.570103 IP 172.16.27.1 > 8.8.8.8: ICMP 172.16.27.1 udp port 58303 unreachable, length 161
```

[Rocky Linux 서버 패킷 분석 - fping]

- **fping** 공격 로그 (ICMP) 확인

fping을 실행했을 때, **tcpdump** 창에 다음과 같은 **ICMP (Ping)** 요청이 Kali IP 주소로부터 들어오는 것이 확인되어야 합니다.

[Kali IP 주소] > [Rocky IP 주소]: ICMP echo request

- 예시: **172.16.27.50 > 172.16.27.154: ICMP echo request**

[arp-scan]

명령어 : arp-scan -l (로컬 서브넷 전체) / arp-scan [CIDR]
172.16.27.0/24 / arp-scan 172.16.27.154

```
(root@kali)-[/home/kalii]
# arp-scan 172.16.27.0/16
Interface: eth0, type: EN10MB, MAC: 00:0c:29:56:44:50, IPv4: 172.16.27.50
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
WARNING: host part of 172.16.27.0/16 is non-zero
Starting arp-scan 1.10.0 with 65536 hosts (https://github.com/royhills/arp-scan)
172.16.0.1      88:36:6c:7a:46:5e      (Unknown)
172.16.0.2      00:90:f5:c6:e4:41      (Unknown)
172.16.0.3      00:11:32:2c:fb:57      (Unknown)
172.16.0.4      00:0c:29:66:7a:ca      (Unknown)
172.16.0.5      68:1d:ef:34:da:66      (Unknown)
172.16.0.6      00:d8:61:79:3a:e7      (Unknown)
172.16.0.7      9c:2d:cd:3f:22:0d      (Unknown)
172.16.27.154  00:0c:29:16:ea:93      (Unknown)
```

IP주소와 MAC주소를 맵핑한 것을 확인할 수 있다.

172.16.27.154 > 00:0c:29:16:ea:93

00:0c:29 = [VMware](#)에서 사용하는 고유한 접두사

로키 명령어 : tcpdump -i ens160 host 172.16.27.50

```
[root@localhost ~]# tcpdump -i ens160 host 172.16.27.50
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens160, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:36:14.969912 ARP, Request who-has 172.16.0.0 tell 172.16.27.50, length 46
16:36:14.972242 ARP, Request who-has _gateway tell 172.16.27.50, length 46
16:36:14.972366 ARP, Reply _gateway is-at 88:36:6c:7a:46:5e (oui Unknown), length
46
16:36:14.974120 ARP, Request who-has 172.16.0.2 tell 172.16.27.50, length 46
16:36:14.974389 ARP, Reply 172.16.0.2 is-at 00:90:f5:c6:e4:41 (oui Unknown), lengt
h 46
16:36:14.975715 ARP, Request who-has 172.16.0.3 tell 172.16.27.50, length 46
16:36:14.975954 ARP, Reply 172.16.0.3 is-at 00:11:32:2c:fb:57 (oui Unknown), lengt
h 46
16:36:14.977691 ARP, Request who-has 172.16.0.4 tell 172.16.27.50, length 46
16:36:14.979265 ARP, Reply 172.16.0.4 is-at 00:0c:29:66:7a:ca (oui Unknown), lengt
h 46
16:36:14.979439 ARP, Request who-has 172.16.0.5 tell 172.16.27.50, length 46
16:36:14.979990 ARP, Reply 172.16.0.5 is-at 68:1d:ef:34:da:66 (oui Unknown), lengt
h 46
16:36:14.981722 ARP, Request who-has 172.16.0.6 tell 172.16.27.50, length 46
16:36:14.981911 ARP, Reply 172.16.0.6 is-at 00:d8:61:79:3a:e7 (oui Unknown), lengt
h 46
16:36:14.984209 ARP, Request who-has 172.16.0.7 tell 172.16.27.50, length 46
16:36:14.985119 ARP, Reply 172.16.0.7 is-at 9c:2d:cd:3f:22:0d (oui Unknown), lengt
h 46
```

[Rocky Linux 서버 패킷 분석 - arp-scan]

16:36:14.974120 ARP, Request who-has 172.16.0.2 tell

172.16.27.50, length 46

16:36:14.974389 ARP, Reply 172.16.0.2 is-at 00:90:f5:c6:e4:41 (oui Unknown), length 46

[netdiscover]

명령어 : netdiscover -r 172.16.27.0/24 (-r = 네트워크 대역에 ARP 스캔을 수행)

IP주소와 MAC주소 + ARP 요청이 보내진 횟수, MAC 제조사 이름을 확인할 수 있다.

로키 명령어 : tcpdump -i ens160 host 172.16.27.50

카운트가 왜있을까

```
17:31:44.188332 ARP, Request who-has 172.16.27.148 (Broadcast) tell 172.16.27.1, length 46
17:31:44.189946 ARP, Request who-has 172.16.27.149 (Broadcast) tell 172.16.27.1, length 46
17:31:44.191338 ARP, Request who-has 172.16.27.150 (Broadcast) tell 172.16.27.1, length 46
17:31:44.192872 ARP, Request who-has 172.16.27.151 (Broadcast) tell 172.16.27.1, length 46
17:31:44.194341 ARP, Request who-has 172.16.27.152 (Broadcast) tell 172.16.27.1, length 46
17:31:44.195897 ARP, Request who-has 172.16.27.153 (Broadcast) tell 172.16.27.1, length 46
17:31:44.197332 ARP, Request who-has 172.16.27.154 (Broadcast) tell 172.16.27.1, length 46
17:31:44.197343 ARP, Reply 172.16.27.154 is-at 00:0c:29:16:ea:93, length 28
17:31:44.198853 ARP, Request who-has 172.16.27.155 (Broadcast) tell 172.16.27.1, length 46
17:31:44.200329 ARP, Request who-has 172.16.27.156 (Broadcast) tell 172.16.27.1, length 46
17:31:44.201841 ARP, Request who-has 172.16.27.157 (Broadcast) tell 172.16.27.1, length 46
17:31:44.203322 ARP, Request who-has 172.16.27.158 (Broadcast) tell 172.16.27.1, length 46
17:31:44.204818 ARP, Request who-has 172.16.27.159 (Broadcast) tell 172.16.27.1, length 46
17:31:44.206242 ARP, Request who-has 172.16.27.160 (Broadcast) tell 172.16.27.1, length 46
17:31:44.207854 ARP, Request who-has 172.16.27.161 (Broadcast) tell 172.16.27.1, length 46
17:31:44.209327 ARP, Request who-has 172.16.27.162 (Broadcast) tell 172.16.27.1, length 46
17:31:44.210816 ARP, Request who-has 172.16.27.163 (Broadcast) tell 172.16.27.1, length 46
```

[Rocky Linux 서버 패킷 분석 - netdiscover]

- arp-scan 및 netdiscover 공격 로그 (ARP) 확인

arp-scan이나 netdiscover를 실행했을 때, tcpdump 창에 다음과 같은 ARP 요청 브로드캐스트가 확인되어야 합니다.

ARP, Request who-has [특정 IP] tell [Kali IP 주소]

- 예시: ARP, Request who-has 192.168.1.200 tell 192.168.1.100